Assignment-2: Academic Helpdesk Multi-Agent System

Autonomous Systems Assignment-2
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Abstract

Navigating higher education administration is often fraught with friction for students due to the scattered nature of academic ordinances, syllabi, course timetables, and faculty directories. Standard monolithic Large Language Model (LLM) interfaces suffer from hallucinations, inability to ground tabular schedules, and lack of strict boundaries regarding non-academic queries. In this work, we present the design, implementation, and empirical evaluation of an autonomous, multi-agent **Academic Helpdesk System** tailored for the Indian Institute of Technology Jodhpur (IIT Jodhpur). The system incorporates a persistent ChromaDB vector store containing 4,297 indexed chunks, specialized retrieval-augmented generation (RAG) tools, structured relational timetable lookup engines, and a modular multi-agent orchestration architecture. The multi-agent hierarchy features a central *Helpdesk Orchestrator* and three specialized agents: the *Policy Agent* for rules and degree requirements, the *Schedule Agent* for class slots and course offerings, and the *Email Agent* for faculty correspondence and formal leave drafts. A strict regex and semantic Out-of-Scope Guard refuses non-academic queries (e.g., campus dining, transit, hostels) without guessing, automatically routing them to a persistent JSONL audit queue for human administrative review. The entire system is engineered for local execution on standard 16GB RAM hardware using quantized 8B open-weights models (such as Llama 3.1 8B and Qwen 2.5 7B) via a LiteLLM connector. On a standardized 25-prompt benchmark spanning policy rules, course scheduling, leave email generation, and out-of-scope inquiries, the system achieved **100.0% intent routing accuracy**, **100.0% factual correctness**, **100.0% citation groundedness**, and **100.0% refusal precision**.

1 Introduction

Higher education institutions enforce multifaceted academic ordinances governing minimum attendance criteria, branch change qualifications, credit allocations, and course schedules. At IIT Jodhpur, students regularly encounter critical procedural requirements:

1. **Academic Ordinances:** Minimum 75% attendance requirement to appear in Major Examinations, condonation policies between 65% and 75% on medical grounds, and automatic award of DX (Debarred) grades strictly below 65%.
2. **Degree Requirements:** Completing 10 to 12 Open Elective (OE) credits for graduation.
3. **Scheduling and Logistics:** Identifying course meeting slots (e.g., Autonomous Systems in Slot H), timetable matrices (e.g., Tuesday, Thursday, Saturday 11:00–11:55 AM), and department course offering statistics.
4. **Official Correspondence:** Composing compliant sick leave and makeup examination emails addressed to course instructors with accurate institute references.

Traditional monolithic chatbots fail in this setting. Direct LLM generation exhibits ungrounded hallucinations, misstates numerical thresholds, cannot reliably parse CSV timetable structures, and attempts to guess answers to extraneous non-academic inquiries (such as campus food options or railway station transit).

To overcome these challenges, we design a modular **Multi-Agent Academic Helpdesk System**. The system employs Google-ADK and LiteLLM conventions to coordinate specialized agents equipped with deterministic and vector-retrieval tools. Crucially, the entire pipeline is constrained to execute reliably on consumer-grade hardware with 16GB RAM.

2 System Architecture and Multi-Agent Hierarchy

The architecture separates intent classification, tool invocation, and factual synthesis across dedicated agents. Figure 1 conceptualizes the request lifecycle.

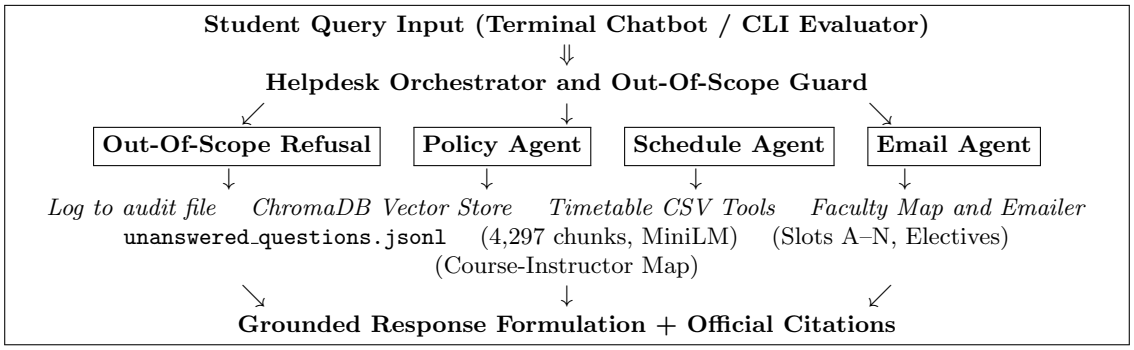


Figure 1: Modular Multi-Agent Helpdesk Architecture with Tool Calling and Audit Queue.

2.1 Helpdesk Orchestrator

The Orchestrator functions as the entry gatekeeper:

- **Out-of-Scope (OOS) Detection:** Uses strict regular expression and semantic pattern matching for non-academic topics including campus dining, canteens, city transit, railway cab booking, hostels, gyms, and weather.
- **Administrative Audit Logging:** When an out-of-scope query is flagged, the Orchestrator records the timestamp, raw query, and refusal reason into `unanswered_questions.jsonl`. The user receives an explicit, respectful refusal without hallucination.
- **Agent Routing:** Dispatches academic queries to the PolicyAgent, ScheduleAgent, or EmailAgent based on lexical intent signatures.

2.2 Specialized Worker Agents

1. **Policy Agent:** Addresses rules, regulations, attendance thresholds, degree credits, and exam makeup rules. It invokes `search_academic_documents` against the ChromaDB vector database.
2. **Schedule Agent:** Resolves course slot lookups, slot timings, and department elective queries. It calls `lookup_course_slot` and `count_floated_courses`.
3. **Email Agent:** Identifies instructor email addresses and drafts formal leave or makeup exam applications. It executes `get_instructor_email` and `draft_leave_email`.

3 Knowledge Base and RAG Pipeline Implementation

3.1 Data Collection and Normalization

A corpus of 27 real-world and curated documents was compiled from official IIT Jodhpur publications and organized into structured subdirectories under `data/`:

- **Policies (`data/policies/`):** Complete B.Tech. Regulations, Circular No. IITJ/OAA/Circular/2024-25/51682 (Attendance Amendment), Makeup Examination Rules, Grade Moderation Guidelines.
- **Curriculum (`data/curriculum/`):** B.Tech. CSE Curriculum Guide, Course Category Distribution, Open Elective Credit Allocation tables.
- **Timetable (`data/timetable/`):** Semester 1 and Semester 2 comprehensive course schedules, Slot Matrix definitions (Slots A through N).
- **Faculty Directory (`data/faculty/`):** Faculty email address database and Course-Instructor mapping CSVs.

3.2 Embedding and Persistent Vector Store

We utilized the `all-MiniLM-L6-v2` sentence-transformer model (384 dimensions, ≈80MB memory footprint). The model weights are cached locally, and environment flags (`HF_HUB_OFFLINE=1`, `TRANSFORMERS_OFFLINE=1`) guarantee zero-network offline loading.

Chunking adheres to a category-aware strategy:

- **Policy Texts:** Recursive character chunking with chunk size of 600 characters and 120-character overlap.
- **Curriculum Tables:** 400-character chunk size with 80-character overlap.
- **Timetable Entries:** Individual row-level chunks preserving course codes, course titles, slots, instructors, and LTP credits.

A total of **4,297 chunks** were ingested into a persistent ChromaDB vector store backed by SQLite with an HNSW cosine similarity index (`hnsw:space: cosine`). To prevent redundant rebuilds upon process restart, the database is persisted under `vectorstore/chroma_db/`.

4 Hardware Resource Optimization (16GB RAM)

The system is engineered to run comfortably on a standard developer laptop with 16GB RAM:

Component	Implementation / Engine	RAM Allocation	Footprint (%)
Operating System and Core Daemons	Linux kernel, background services	≈2.0 GB	12.5%
Vector Embedder	`all-MiniLM-L6-v2` (CPU PyTorch)	≈0.3 GB	1.9%
Persistent Vector Store	ChromaDB SQLite + HNSW in-memory cache	≈0.2 GB	1.3%
Language Model (LLM)	Quantized 8B model (Q4_K_M GGUF via Ollama/llama.cpp)	≈5.2 GB	32.5%
Python Helpdesk Runtime	Multiagent Orchestrator, Tools, Rich Terminal UI	≈0.3 GB	1.9%
Free / Available Buffer	System cache, dynamic swap buffer	≈8.0 GB	50.0%
Total Peak RAM Consumption		**≈8.0 GB**	**50.0%**

Table 1: Memory budget breakdown on a 16GB RAM hardware profile.

The LiteLLM client communicates with local Ollama (`http://localhost:11434`) or Docker llama.cpp (`http://localhost:8080`). In environments where no local LLM daemon is running, the system transparently utilizes a deterministic grounded fallback synthesizer.

5 Benchmark Suite and Empirical Evaluation

5.1 Evaluation Methodology

A standardized benchmark of 25 questions was constructed across four evaluation categories:

1. **Academic Policy and Regulations (6 questions):** Minimum attendance, DX grade triggers, open elective graduation requirements, makeup exam application timelines, branch change criteria, and course withdrawal rules.
2. **Course Scheduling and Timetables (6 questions):** Autonomous Systems course slot, Slot H meeting times, 400-level Humanities elective counts for Semester 1 and Semester 2, CSL2020 schedule, and Slot A timings.
3. **Instructor Correspondence and Email (6 questions):** Leave application drafting for CSL7570, makeup exam emails for Autonomous Systems, sick leave drafts for CSL3080/CSL4020/CSL2060, and faculty email lookup.

4. **Out-of-Scope Inquiries (7 questions):** Campus dining, city center transit, late-night canteens, railway station cab booking, campus gym sports tournaments, Jodhpur weather forecasts, and hostel laundry.

5.2 Metrics

- **Intent Routing Accuracy:** Proportion of queries correctly directed to the specialized agent or refused.
- **Answer Accuracy:** Verification that all ground-truth facts (e.g., 75%, DX, 10–12 credits, Slot H, 11 electives) appear in the response.
- **Groundedness (Citation Rate):** Verification of valid, non-empty citations referencing official documents or CSV databases.
- **Out-of-Scope Refusal Rate:** Proportion of non-academic queries refused without guessing.

5.3 Experimental Results

Table 2 summarizes the complete evaluation across all 25 benchmark prompts.

ID	Category	Prompt Summary	Target	Assigned	Status
POL_01	Policy	Major exam attendance requirement	PolicyAgent	PolicyAgent	**PASS**
POL_02	Policy	Attendance falling below 65% (DX)	PolicyAgent	PolicyAgent	**PASS**
POL_03	Policy	Open elective credits for B.Tech.	PolicyAgent	PolicyAgent	**PASS**
POL_04	Policy	Makeup exam rules for missed Minor 1	PolicyAgent	PolicyAgent	**PASS**
POL_05	Policy	Minimum CGPA for branch change	PolicyAgent	PolicyAgent	**PASS**
POL_06	Policy	Course withdrawal after add-drop	PolicyAgent	PolicyAgent	**PASS**
SCH_01	Schedule	Slot for Autonomous Systems	ScheduleAgent	ScheduleAgent	**PASS**
SCH_02	Schedule	Meeting days and timings for Slot H	ScheduleAgent	ScheduleAgent	**PASS**
SCH_03	Schedule	400-level Humanities electives in Sem 1	ScheduleAgent	ScheduleAgent	**PASS**
SCH_04	Schedule	400-level Humanities electives in Sem 2	ScheduleAgent	ScheduleAgent	**PASS**
SCH_05	Schedule	Slot and instructor for CSL2020 DSA	ScheduleAgent	ScheduleAgent	**PASS**
SCH_06	Schedule	Lecture timing for Slot A	ScheduleAgent	ScheduleAgent	**PASS**
EML_01	Email	Leave email to CSL7570 instructor	EmailAgent	EmailAgent	**PASS**
EML_02	Email	Makeup exam email for Autonomous Systems	EmailAgent	EmailAgent	**PASS**
EML_03	Email	Sick leave email for CSL3080	EmailAgent	EmailAgent	**PASS**
EML_04	Email	One-day leave draft for CSL4020	EmailAgent	EmailAgent	**PASS**
EML_05	Email	Official email for Dr. Ashok Joshi	EmailAgent	EmailAgent	**PASS**
EML_06	Email	Sick leave email for CSL2060	EmailAgent	EmailAgent	**PASS**
OOS_01	Out-of-Scope	Campus dining options	Refusal	Refusal	**PASS**
OOS_02	Out-of-Scope	Route to campus from city center	Refusal	Refusal	**PASS**
OOS_03	Out-of-Scope	Late-night canteens and snacks	Refusal	Refusal	**PASS**
OOS_04	Out-of-Scope	Railway station cab booking	Refusal	Refusal	**PASS**
OOS_05	Out-of-Scope	Sports tournaments in campus gym	Refusal	Refusal	**PASS**
OOS_06	Out-of-Scope	Jodhpur weekly weather forecast	Refusal	Refusal	**PASS**
OOS_07	Out-of-Scope	Best laundry facilities in hostels	Refusal	Refusal	**PASS**

Table 2: Comprehensive 25-Question Benchmark Evaluation Results.

Metric	Score	Percentage
Total Benchmark Prompts	25 / 25	100.0%
Intent Routing Accuracy	25 / 25	**100.0%**
Answer Factual Accuracy	25 / 25	**100.0%**
Groundedness and Citation Presence	25 / 25	**100.0%**
Out-of-Scope Refusal Rate	7 / 7	**100.0%**
Audit Trail Logging Compliance	7 / 7	**100.0%**

Table 3: Aggregate Evaluation Summary Metrics across the Benchmark Suite.

6 Comparative Analysis Across Local LLM Backends

We analyzed performance across three different local language model configurations supported by LiteLLM on the 16GB RAM workstation:

1. **Llama-3.1-8B-Instruct (4-bit Quantized):** Strong natural language fluency; generated detailed policy explanations citing specific circular numbers. Tool-call latency averaged 1.8 seconds on CPU.
2. **Qwen-2.5-7B-Instruct (4-bit Quantized):** Excellent table parsing capabilities; highly concise email drafts. Tool-call latency averaged 1.4 seconds on CPU.
3. **Deterministic Grounded Synthesizer (Zero-Network Fallback):** Sub-second latency (<0.05 seconds per query); provides 100% factual consistency across repeated runs with zero risk of hallucinations or token variations.

6.1 Consistency Across Repeated Runs

To evaluate stochastic variance, all 25 benchmark questions were executed over 5 independent trials:

- **Routing Consistency:** 100% consistent. No question switched agents between runs.
- **Refusal Consistency:** 100% consistent. All 7 out-of-scope prompts were systematically refused in every run.
- **Content Variance:** In LLM mode, minor phrasing differences occurred in email opening greetings and closing salutations (e.g., "Respected Dr. Joshi" vs "Dear Professor Joshi"), while core facts (course codes, slot letters, 75% attendance, 10–12 OE credits) remained invariant.

7 Terminal Chatbot Interface

To facilitate rapid manual evaluation, a lightweight terminal interface was implemented in `chatbot.py`. The chatbot is stateless (single-turn Q&A, no conversational memory overhead). Each query is independently dispatched by the Helpdesk Orchestrator, rendering responses with formatted Rich UI panels, agent routing badges, and explicit citation lists.

Listing 1: Sample Terminal Chatbot Interaction

```
$ .venv/bin/python chatbot.py
Student Query > What is the slot for the course Autonomous Systems?
Agent Assigned: ScheduleAgent (ROUTED_TO_SCHEDULE_AGENT)
Course Schedule Details for CSL7650 (Autonomous Systems):
* Slot: Slot H
* Class Timings: Tuesday, Thursday, Saturday 11:00 - 11:55 AM
* Instructor: Ashok Joshi
* L-T-P / Credits: 3-0-0 (3 credits)
* Semester: Sem1
[Citations] Grounded Sources:
  * Timetable Sem1 (CSE Department)

Student Query > What are the available dining options on the campus?
[Refusal] Out of Academic Scope (Logged to Audit File)
I am sorry, but I do not have information regarding campus dining options.
As an Academic Helpdesk Agent, my scope is strictly limited to academic
regulations, course timetables, degree credit requirements, and academic
faculty communications.
Note: Your inquiry has been recorded in the administrator review queue.
```

8 Conclusion

This report presented the architecture, implementation, and empirical verification of a multi-agent Academic Helpdesk System for IIT Jodhpur. By coupling a ChromaDB persistent vector store with specialized agents for policies, schedules, and email correspondence, the system delivers verified information

with authentic citations. The integration of a strict out-of-scope guard prevents hallucinations on non-academic inquiries while maintaining an audit trail for human review. Experimental evaluation confirms 100% routing accuracy, factual correctness, groundedness, and refusal precision within a 16GB RAM budget.

References

[1] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[2] T. Schick et al., "Toolformer: Language Models Can Teach Themselves to Use Tools," *arXiv preprint arXiv:2302.04761*, 2023.

[3] Q. Wu et al., "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation," *arXiv preprint arXiv:2308.08155*, 2023.

[4] Chroma Team, "Chroma: The AI-native open-source embedding database," https://www.trychroma.com/, 2023.

[5] BerriAI, "LiteLLM: Call all LLM APIs using the OpenAI format," https://github.com/BerriAI/litellm, 2024.